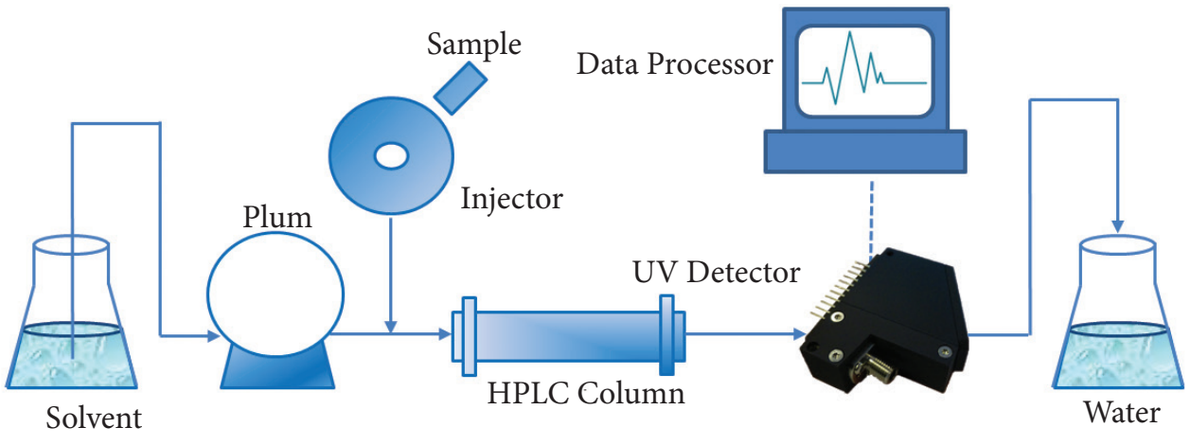


Figure 10.8 : High performance liquid chromatography

10.4. Electrophoresis

In 1948, a Swedish Physical Biochemist, Arne Tiselius was awarded the Nobel prize in chemistry for the discovery of proteins in blood serum and for studying the properties of proteins through electrophoresis. Till now, electrophoresis continues to be an important technique to identify and characterize biological macromolecules. Amino acids, peptides, proteins and nucleic acids possess ionisable groups and can be made to exist in solution as cations or anions. When a mixture of these components is subjected to an electric field, they migrate differently and can be separated.

Principle

Electrophoresis is defined as the migration of charged particles, under the influence of an electric field at a definite pH. In a mixture of proteins, each protein with its electrical charge will move differently in an electric field. This electrophoretic mobility depends on the pH of the medium, strength of the field, net charge of the molecule and size/shape of the molecule. Electrophoresis is used for the analysis of large molecules (proteins and nucleic acids) and simpler charged molecules (peptide, simpler ions).

Types of electrophoresis

The following are different types or electrophoresis.

1. Paper electrophoresis
2. Cellulose acetate electrophoresis
3. Capillary electrophoresis
4. Gel electrophoresis

Agarose gel electrophoresis, Polyacrylamide Gel Electrophoresis (SDS PAGE, Native PAGE and two- dimensional electrophoresis).

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